



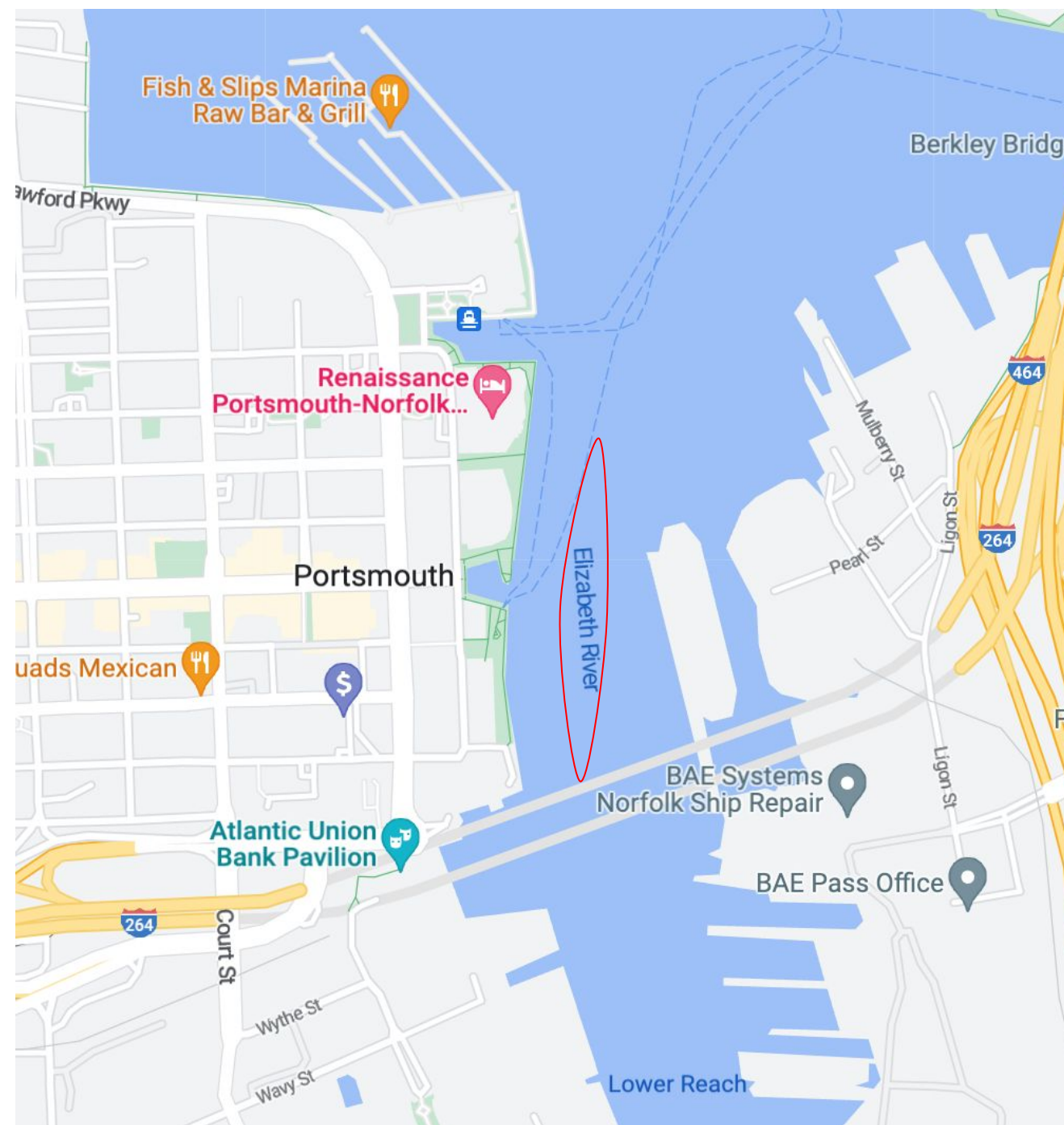
ASNE23 - PEP Electric Boat Competition



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Project Background

We are participating in this year's annual Promoting Electric Propulsion (PEP) competition hosted by the American Society of Naval Engineers (ASNE). To inspire the creation of eco-friendly alternatives, this competition challenges student groups to design and manufacture fully electric-powered boats. The boats will compete in a five-mile race that will be held in Portsmouth, Virginia this June.

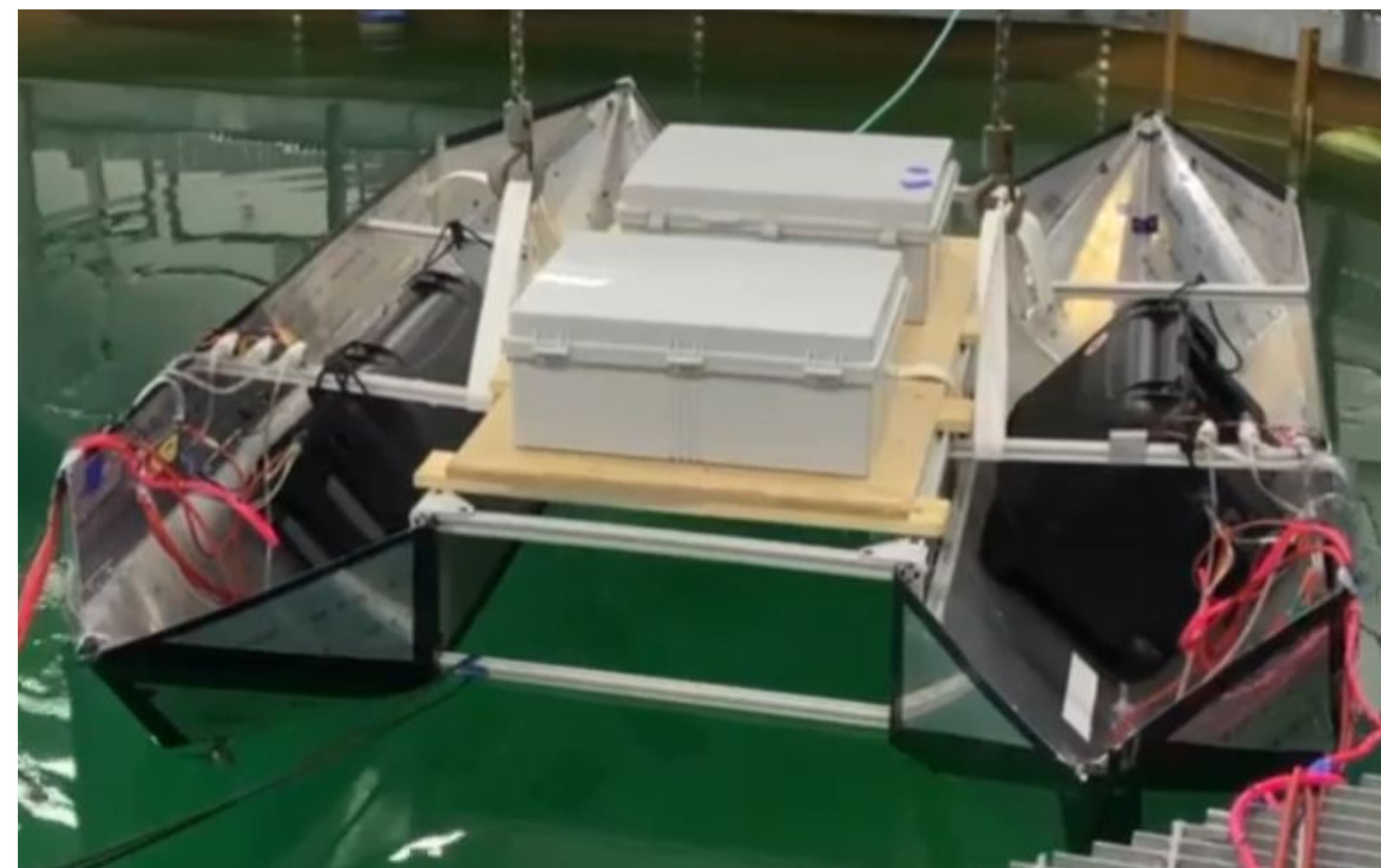


Design Objective

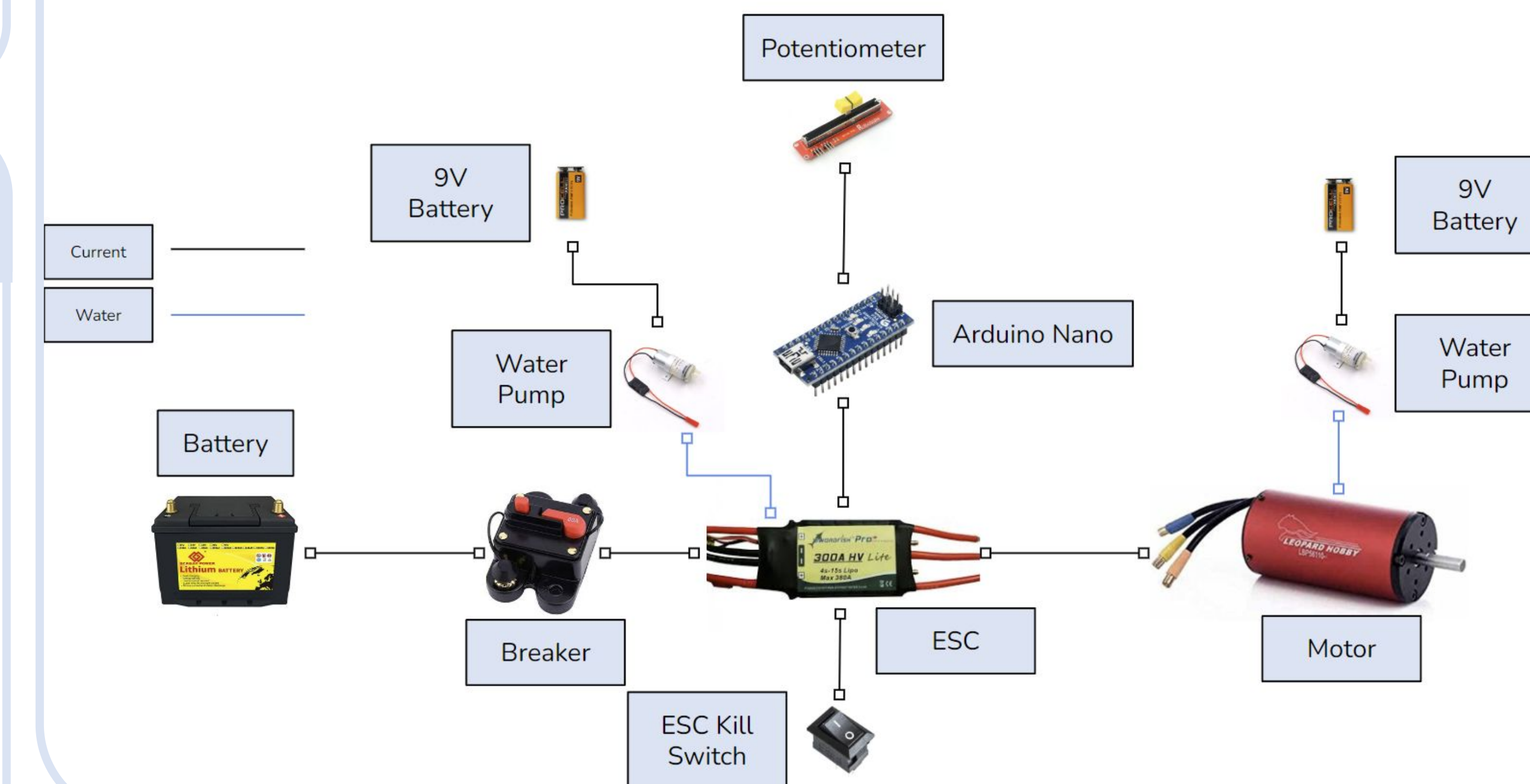
ASNE23 aims to submit a human piloted entry to this year's competition capable of completing the race course in record time.

Design Overview

Our design is a low drag catamaran-style hull design that allows for high stability. The propulsion system is composed of two high-power 7600W brushless DC motors powered by 3 LiPoFe batteries wired in series. This system is connected to our custom made powertrains. An ergonomic driver cell allows for a low center of gravity, and the dual motors allows sharp turning without the added drag and complexity of a rudder.



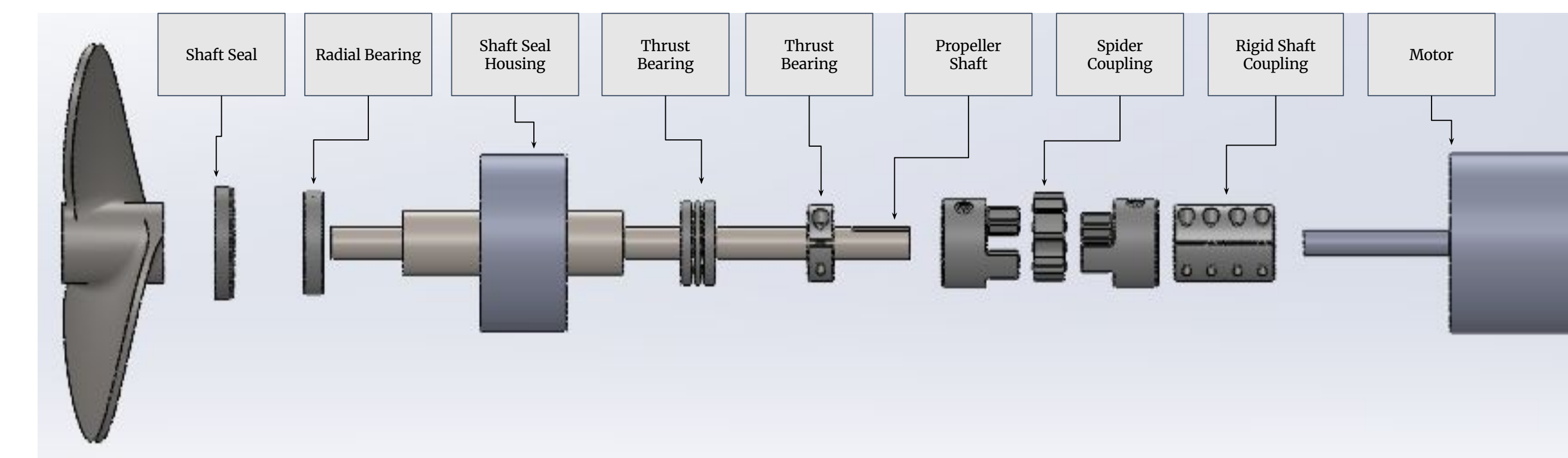
Electronics Overview



Design Requirements

- Must complete the five mile course in under 40 minutes
- Cannot be powered by gasoline engines, sails, or manual propulsion
- Design must comply with USCG safety regulations
- Operated by human pilot

Powertrain Exploded View



Testing, Optimization, and Evaluation

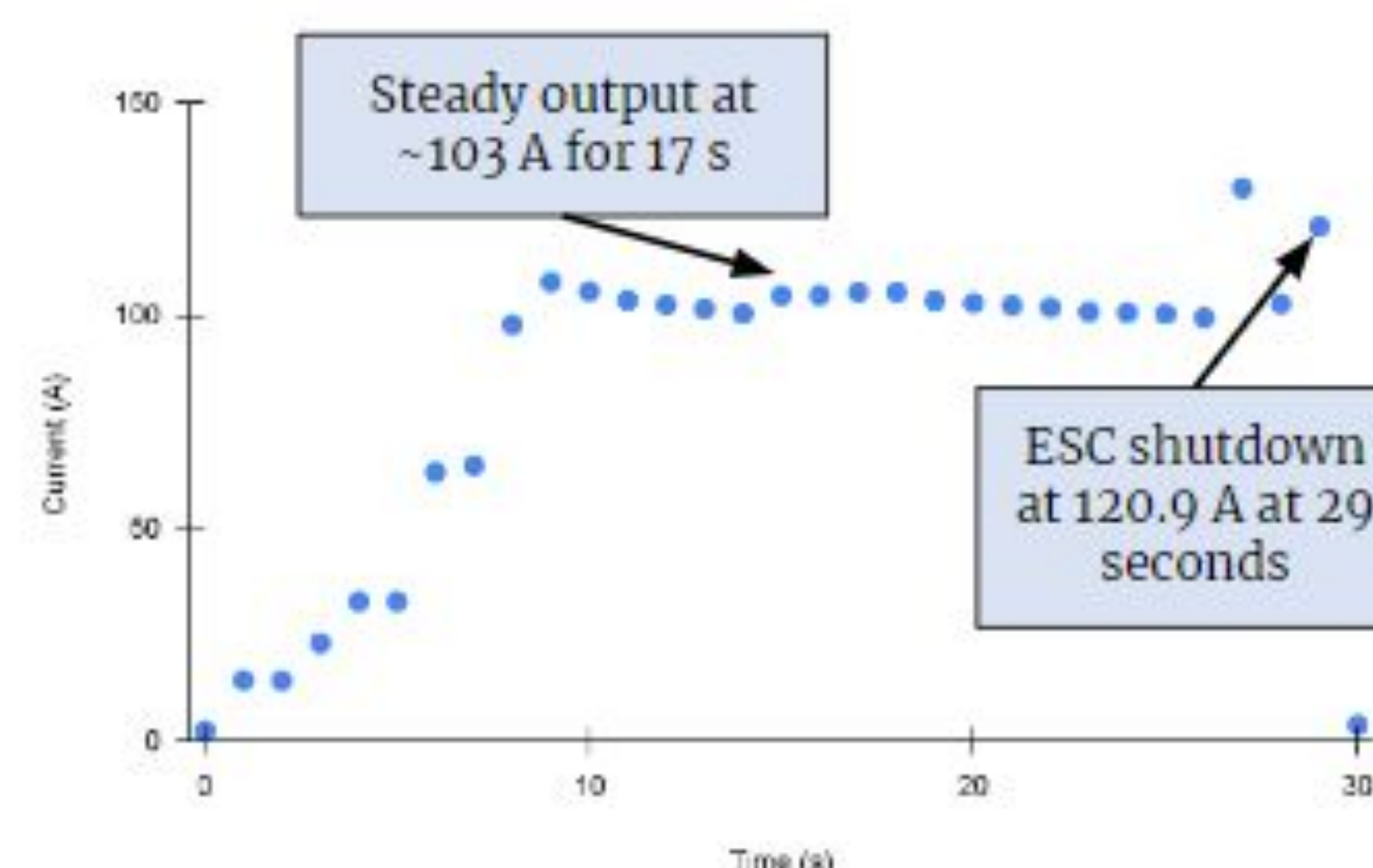
Prolonged "Dry" Test:

One hull was propped over a storm drain and the shaft seal was sprayed with water as it ran at 20% power for 22 minutes. This provided cooling and lubrication for the rubber shaft seal outside of the tank for an extended time. The motor reached 145 °F so the test was stopped. Watercooling was implemented for the motors and ESC to address this.



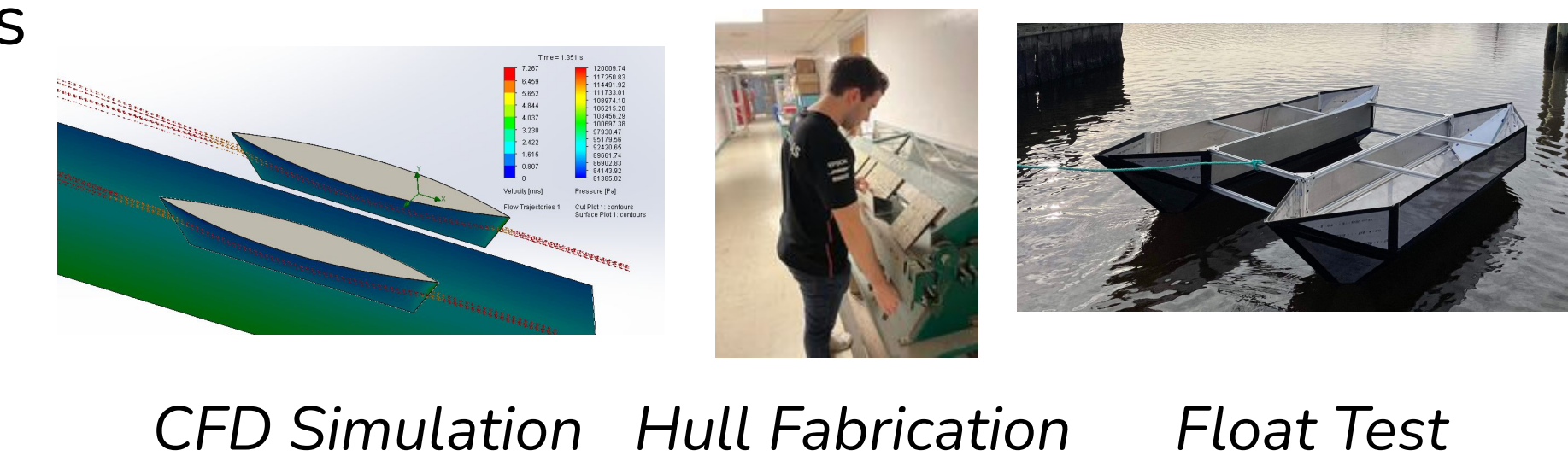
In-Water Test:

The boat was lifted into Professor Whitcomb's ROV tank and tethered from the sides and rear. Current draw measurements were taken over a 30 second test., Motor and ESC temperatures were constantly monitored. Following this test, we are preparing the boat for a full 5-mile in water test at Merritt Point Park.



Design Process

- Brainstorm Solutions
- Model and Simulate
- Manufacture
- Test
- Identify Problems
- Repeat



Next Steps

- Modify boat design based on ROV tank test results
- Test design by piloting the boat for a 5-mile test at Merritt Point Park
- Make any necessary final improvements
- Compete!

Acknowledgements

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